

Toxic Comment Analysis

Deep Learning & Big Data

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About Dataset

- Dataset taken from Kaggle's **Toxic Comment Classification Challenge**
- It contains Wikipedia comments labelled by human raters for toxic behaviour.
- The goal is to predict the probability of each toxicity label for every comment.

Motivation for choosing this toxic comment classification

Motivation

- Online comment sections can contain useful discussion and harmful language at the same time.
- Manual review is expensive when platforms receive many comments.
- To explore how machine learning methods can be used for online content moderation.

Reason for machine learning

Machine learning was used because the task requires recognising text patterns across many comments, not just checking a fixed list of words.

■ The Data set description

The dataset comes from Kaggle: <https://www.kaggle.com/c/jigsaw-toxic-comment-classification-challenge>. The project contains three data files as described below in the table

File	Rows	Columns	Purpose
train.csv	159,571	8	Training comments with text and six toxicity labels
test.csv	153,164 (63,978 usable)	2	Test comments with IDs and comment text without labels
test_labels.csv	153,164 (63,978 usable)	7	Labels for the test comments

`test.csv` has **153,164 comments**, but `test_labels.csv` contains **89,186 rows where the labels are -1**. In the Kaggle dataset, -1 means those rows were **not scored / not usable for evaluation**. Since a confusion matrix needs the true label to know whether each prediction is right or wrong, those rows are removed.

The training data is clean, and there are no missing values

There are no missing values or duplicate IDs.

0

Missing rows

0

Duplicates

63,978

Usable Test comment

The training data is clean, but strongly imbalanced.

First-stage exploration of the Jigsaw toxic comment training data.

159,571

training comments

Rows loaded from train.csv

6

toxicity labels

A comment can have more than one label

10.17%

Unclean comment

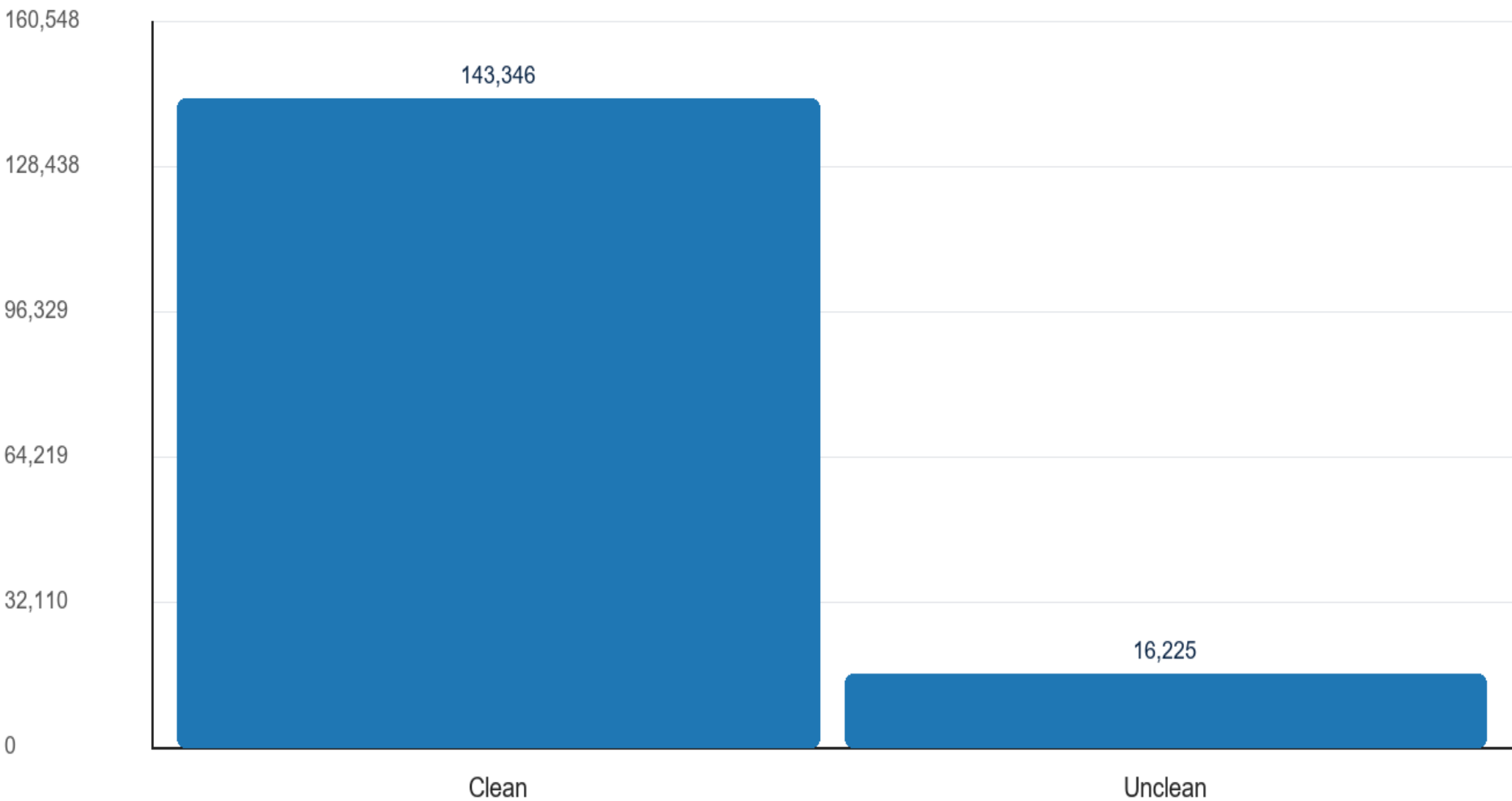
Most training comments are clean

Most training comments are clean.

For the binary task, any comment with at least one toxicity label is counted as toxic.

Clean Comments vs. Unclean Comments

Only 10.17% of training comments have at least one toxicity label.

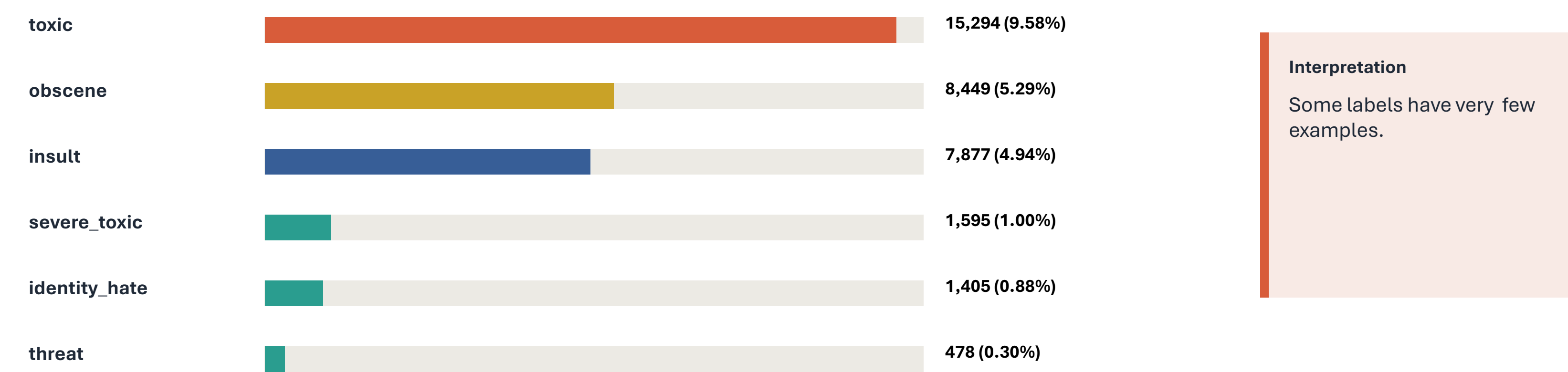


143,346
clean comments 89.83%
All six labels are 0

16,225
unclean comments 10.17%
At least one label is 1
some comments have multiple labels.

Toxicity labels are rare, especially threat and identity hate.

The model must be judged with more than accuracy because the toxic class are small.



Methodology

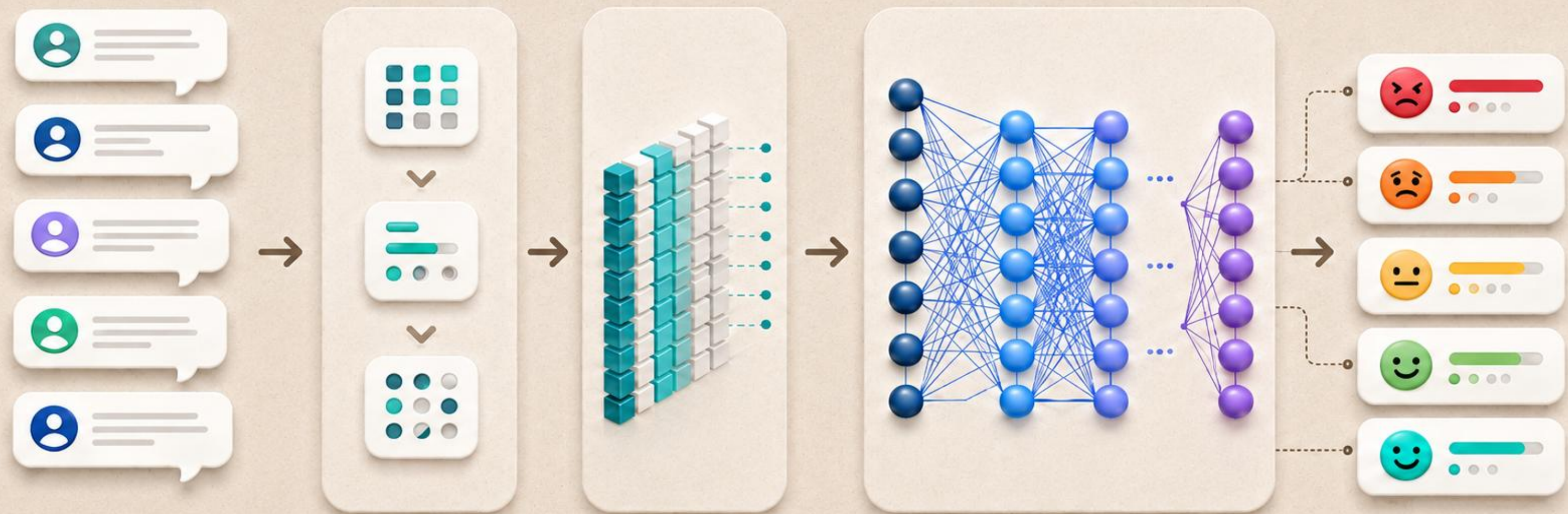
Tasks:

- Multi-Label classification
- Binary classification

Models:

- Neural Network from Scratch
- FineTuned DistilBert Model

Multi-Label Classification



Multi-Label Classification (From Scratch)

- Goal:**
- Train neural network from scratch.
 - Learn

Input and Preprocessing

Text cleaning:

- Lowercase

Aa

- Remove special characters

~~@ # \$ % ^ & * () # !~~

- Remove extra spaces

Word1_ _ Word2

Input and Preprocessing

Tokenization:

Comment: "You are stupid"

Tokens: ["you", "are", "stupid"]

Token IDs: [2017, 2024, 5236]



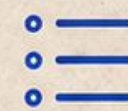
Total tokens

8,833,522



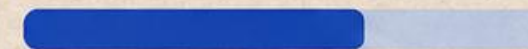
Unique words

160,413

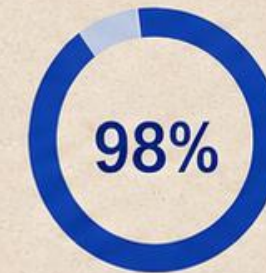


Vocabulary used

50,000



Selected vocabulary size



Token coverage

98%

Input and Preprocessing

Input Length and Padding:

- Padding makes all comments the same length
- Short comments are padded
- Long comments are shortened



Minimum length

1



Maximum length

1403



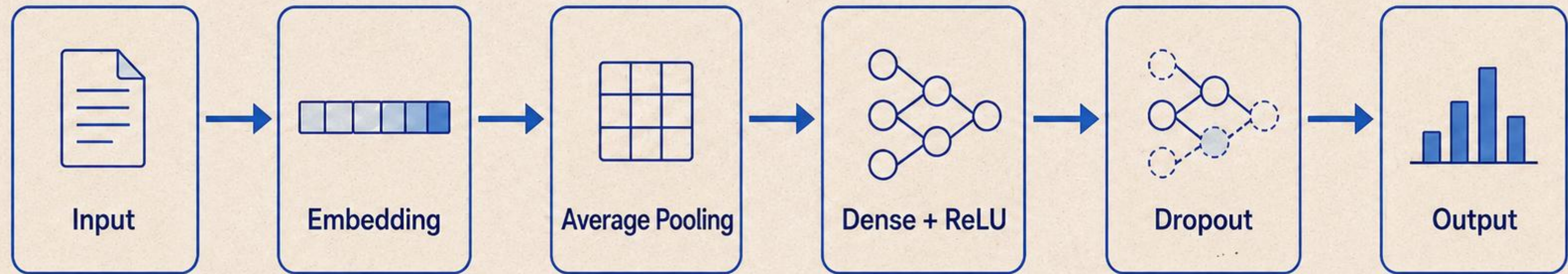
90th Percentile length

165



used value

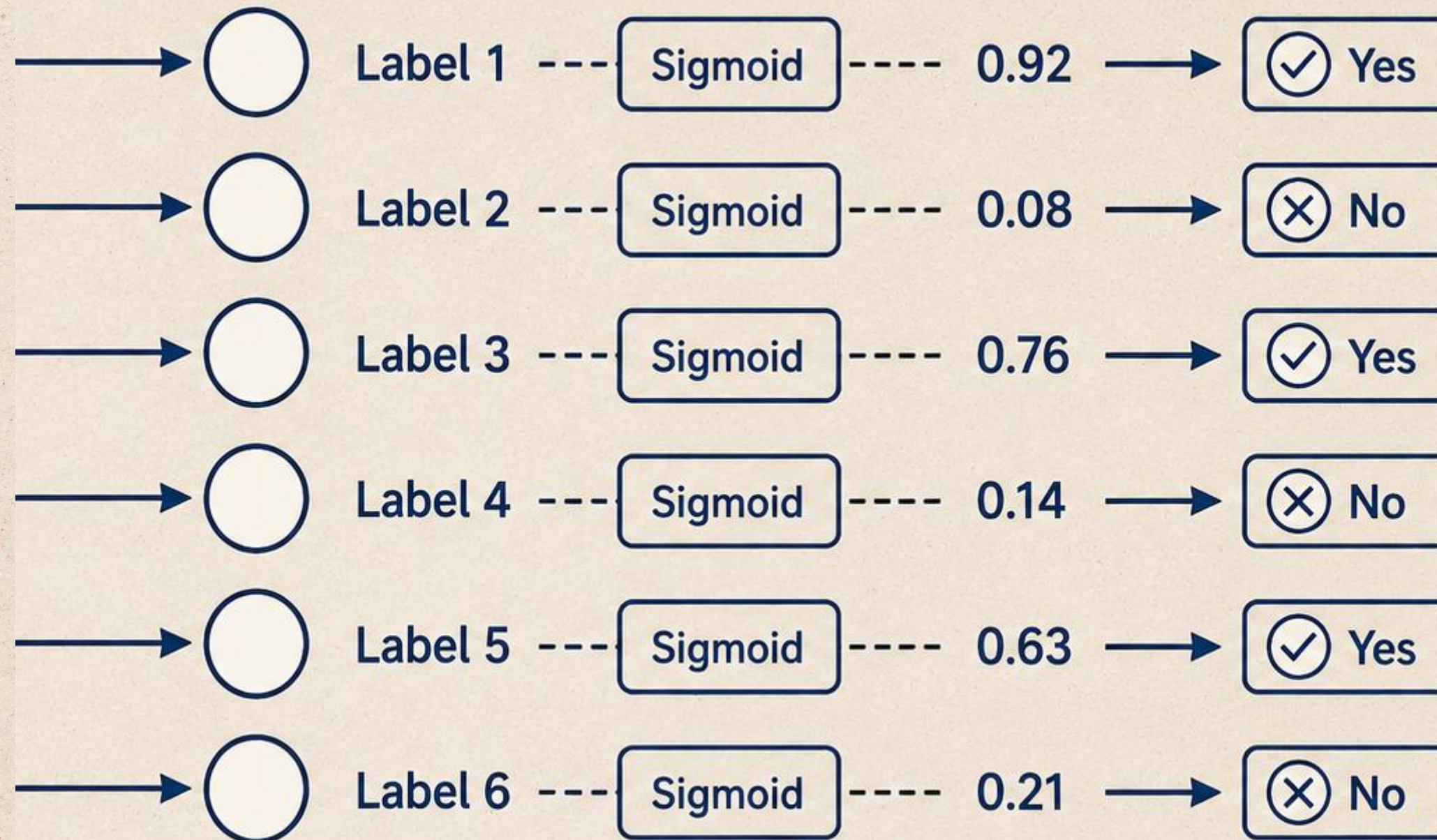
Model Architecture



- Text is converted into embeddings
- Summarized by average pooling
- Passed through a dense layer with ReLU activation, regularized by dropout
- Output layer: 6 labels (Sigmoid function)

Model Architecture

Output layer: 6 labels

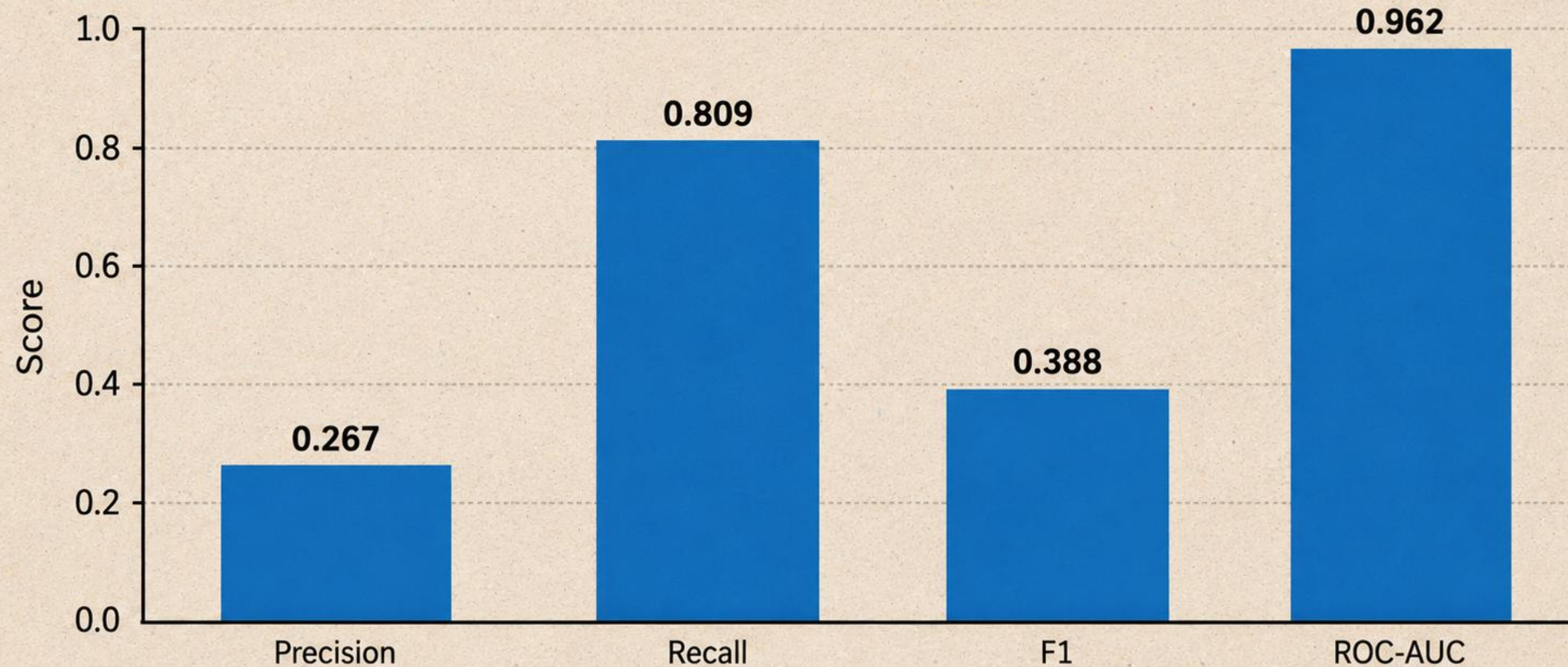


Training Setup

- Loss: BCEWithLogitsLoss
- Optimizer: Adam
- Class weights used for rare labels

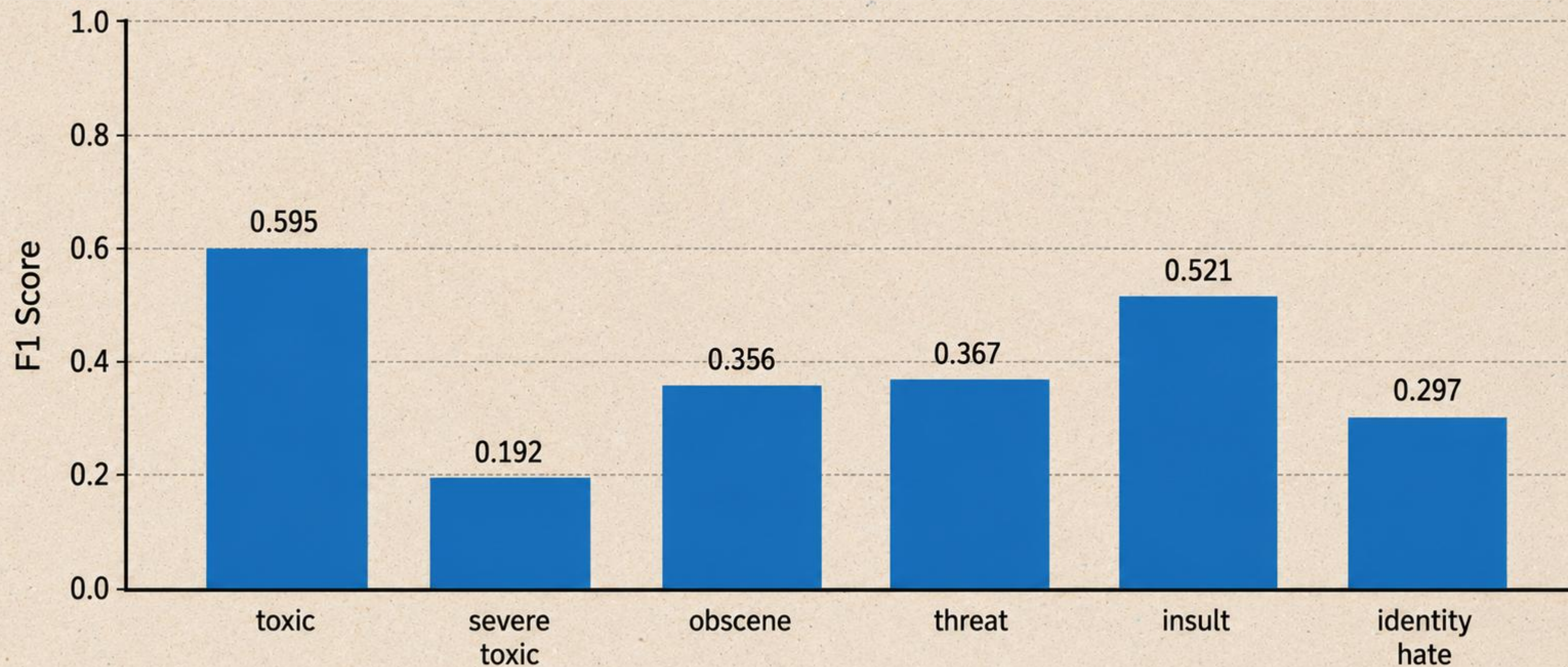
Results

From Scratch Model - Overall Scores



Results

From Scratch Model - F1 by Label

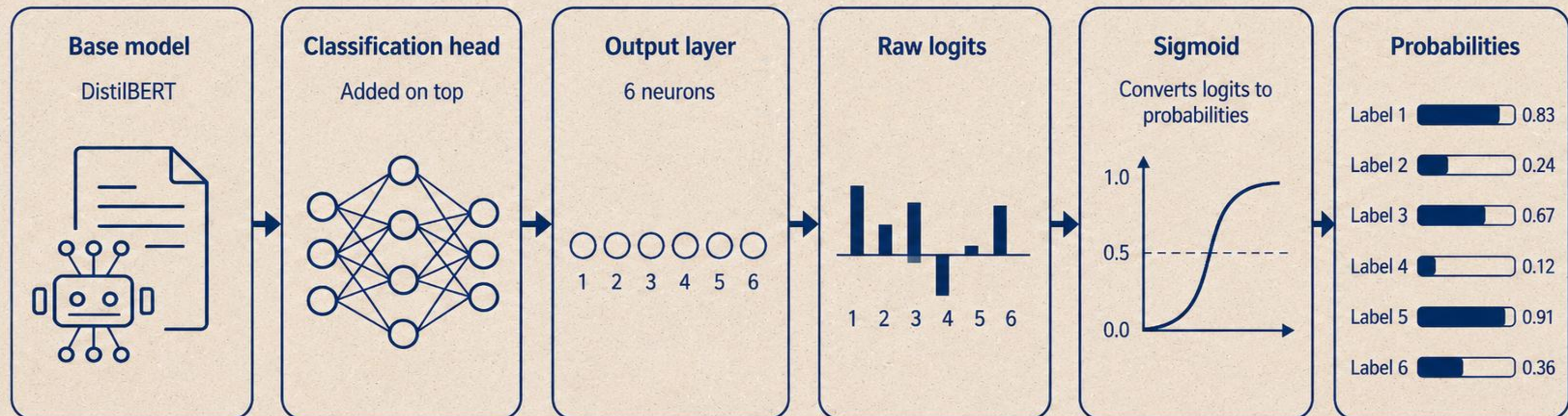


Multi-Label Classification (Transfer Learning)

- Used pretrained DistilBERT
- Fine-tuned for toxic comment classification
- Why DistilBERT?
 - Suitable for multi-label transfer learning
 - Lightweight to train on Laptop / Colab
 - Better than LSTM / CNN

Model Architecture

- Base model: DistilBERT
- Classification head added on top
- Output layer: 6 neurons
 - One neuron for each label
 - Output: raw logits

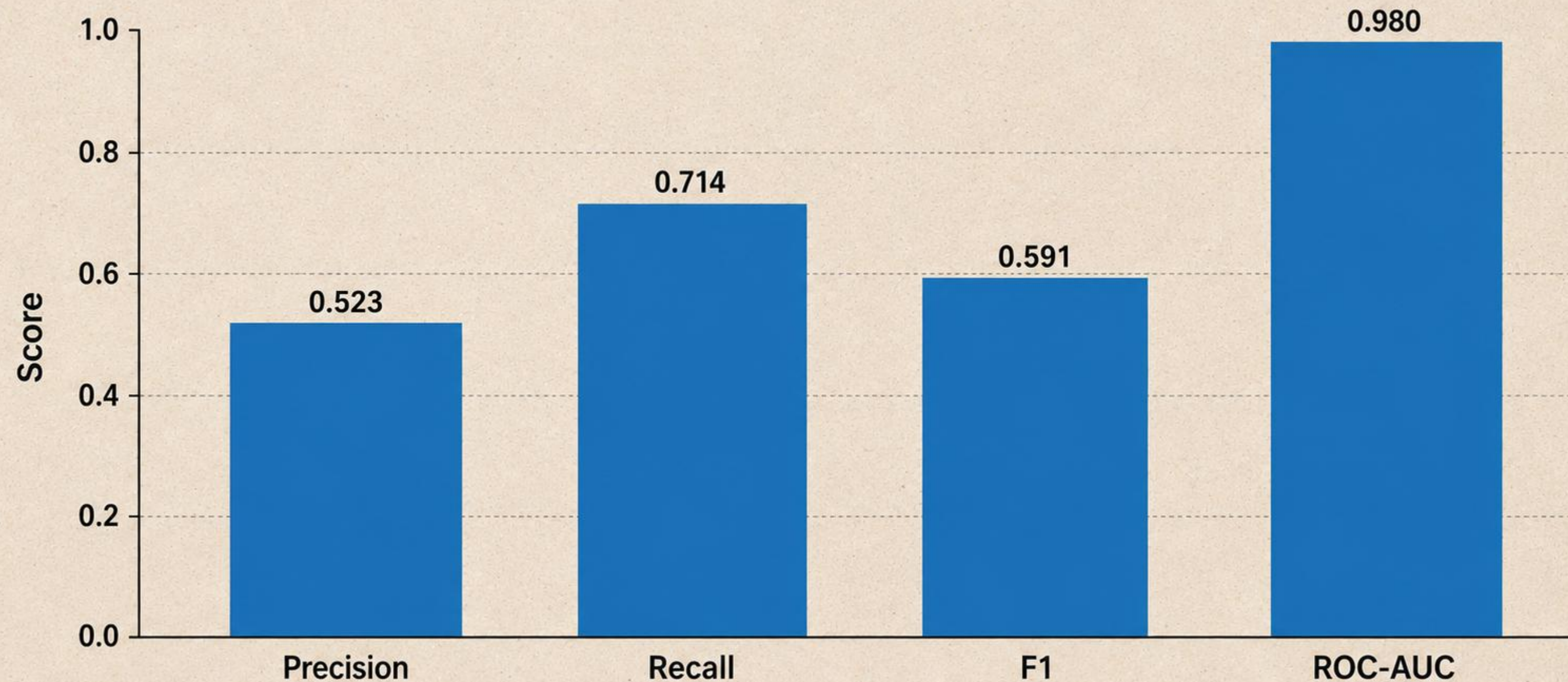


Training Setup

- Epochs: 2
- Batch size: 16
- Evaluation after each epoch
- Best model selected using ROC-AUC
- Trainer: Hugging Face Trainer

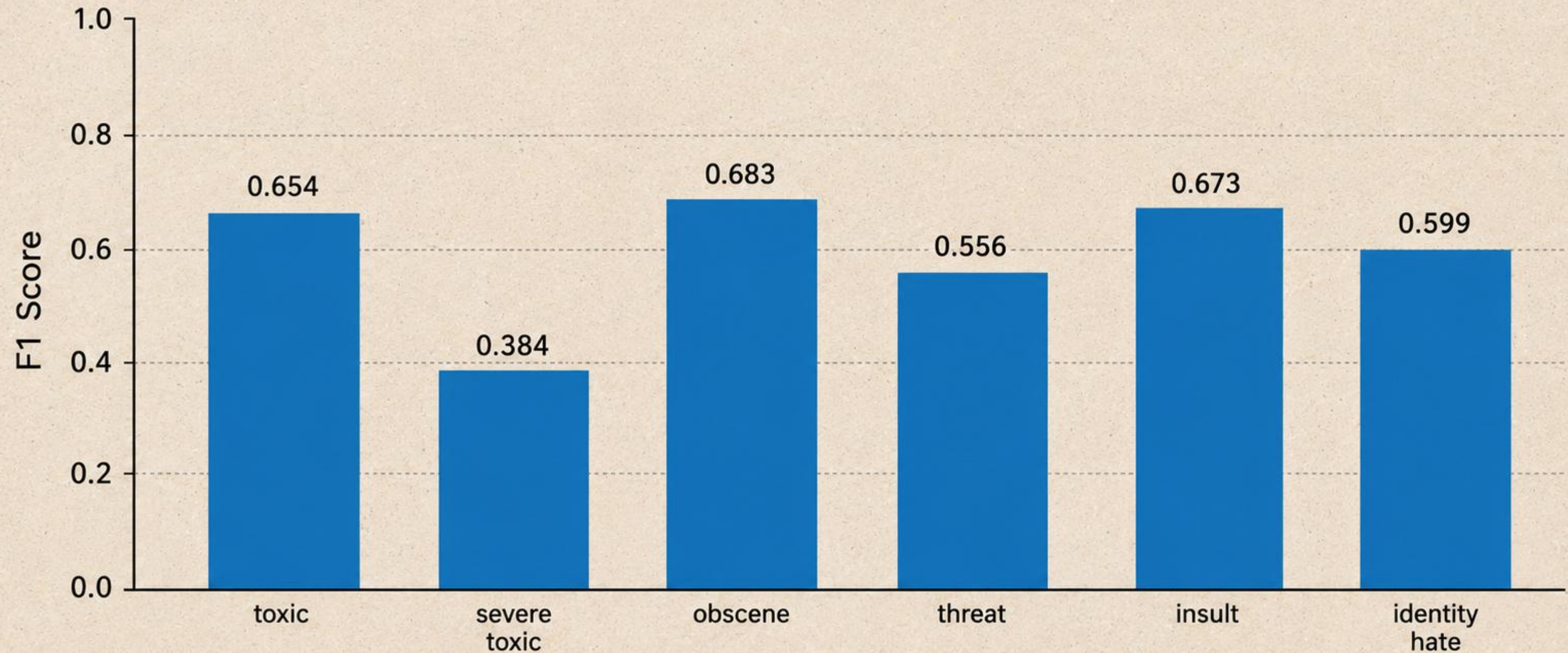
Transfer Learning

Overall Results



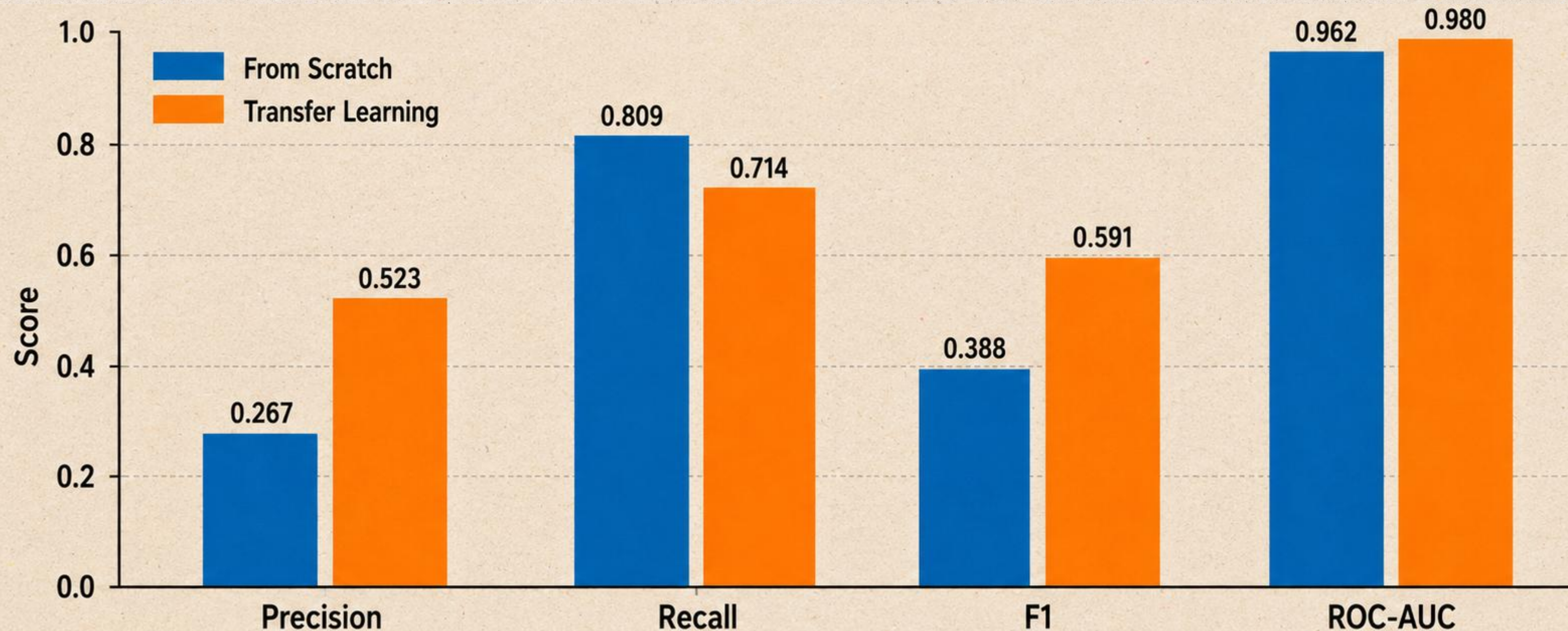
Transfer Learning

Results: F1 by Label



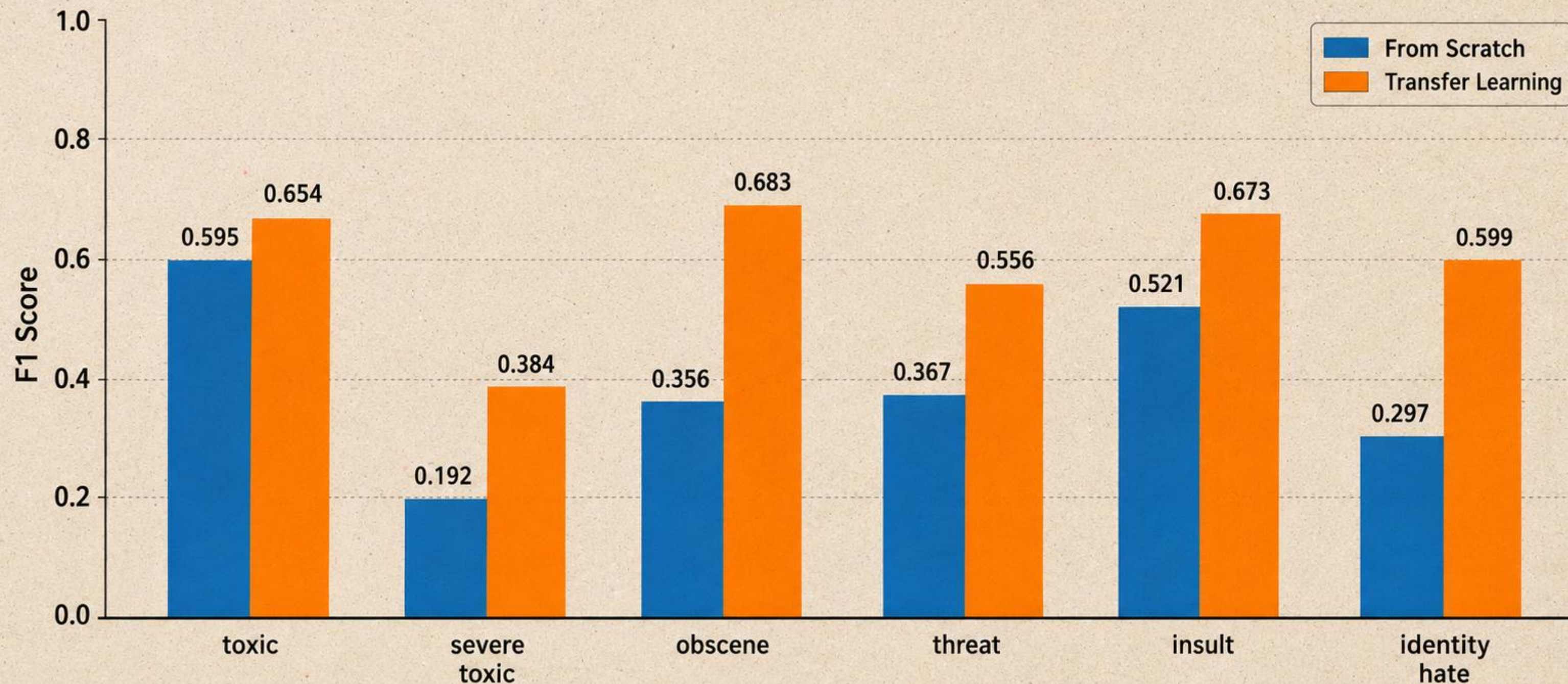
Transfer Learning vs From Scratch

Results: Overall Comparison



Transfer Learning vs From Scratch

Results: F1 Score by Label



Binary Classification

The Labels Are grouped into one single class

How the binary label is made

■ clean = all six toxicity labels are 0

■ unclean = at least one toxicity label is 1

This turns the original multi-label problem into a binary task.

Binary Task and Data Split

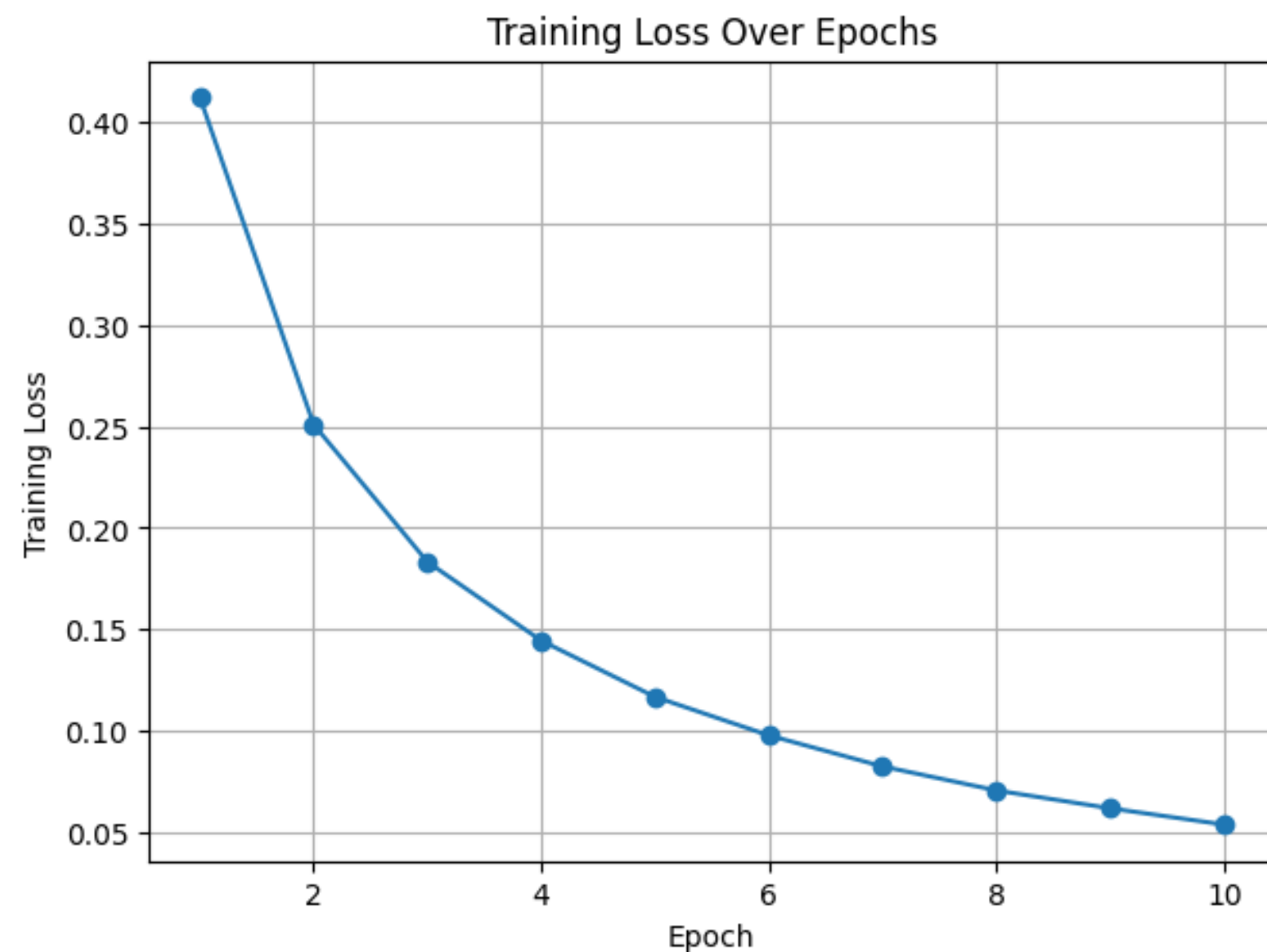
Training rows 159,571	Train split 127,656	Validation split 31,915	Usable test rows from the original 153,164 63,978
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Why accuracy is not enough

- Only about 10.17% of training comments are unclean.
- The comparison focuses on AUC, precision, recall, F1, and confusion matrices.

Binary NN from scratch Classification



Training

The model was trained for 10 epochs

The batch size was: 128

Metrics	Values
Threshold	0.75
Test AUC	0.948
Test accuracy	0.912
Test precision	0.533
Test recall	0.808
Test F1	0.642

Binary NN from scratch Classification

Binary NN Test Confusion Matrix

	Predicted clean	Predicted unclean
Actual clean	53,310	4,425
Actual unclean	1,201	5,042

Green cells are correct decisions. Yellow/red cells are review errors.

Binary NN Validation Confusion Matrix

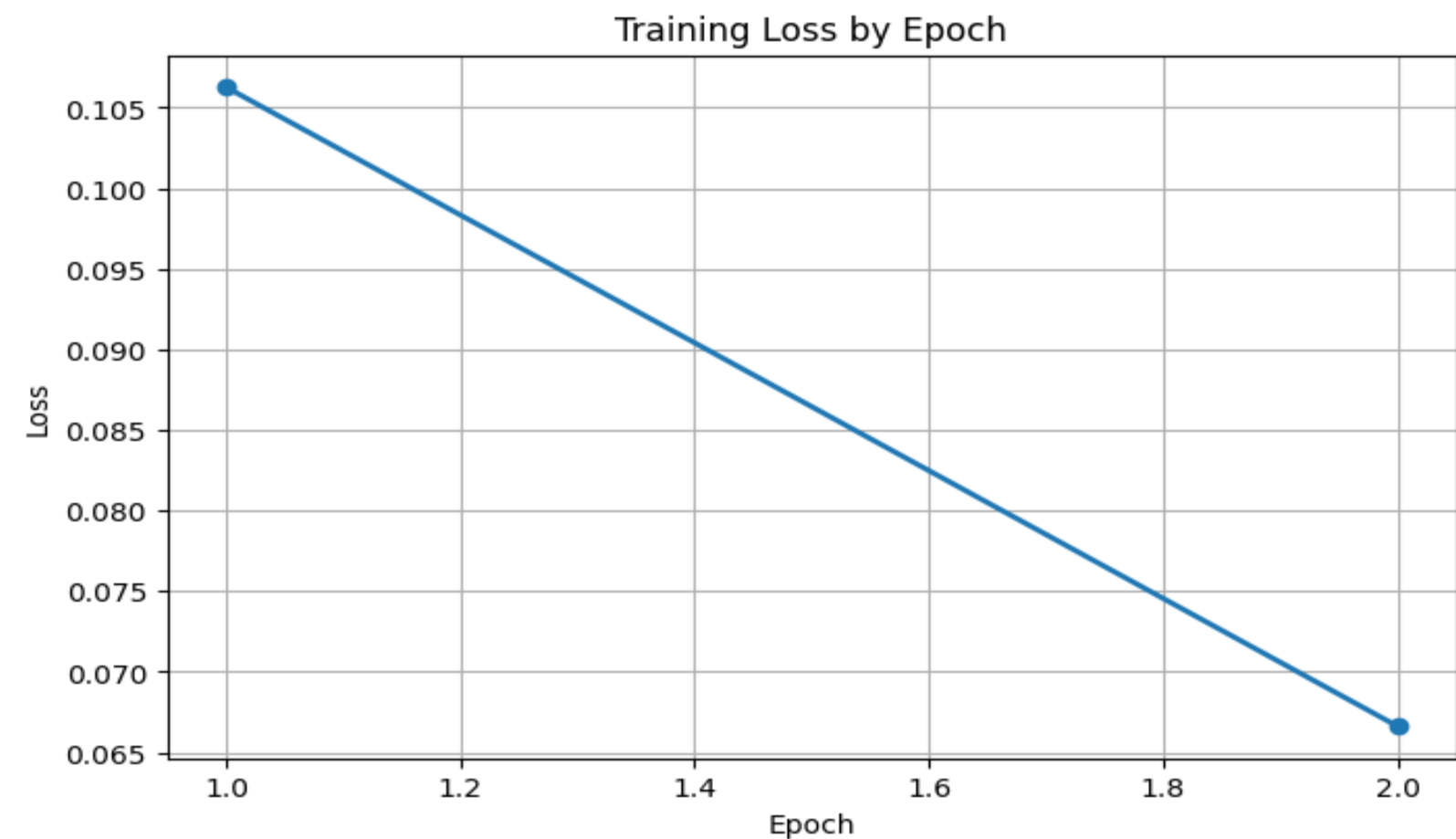
	Predicted clean	Predicted unclean
Actual clean	28,177	493
Actual unclean	929	2,316

Green cells are correct decisions. Yellow/red cells are review errors.

The model performs well overall, especially at identifying Clean comments, but it is weaker at detecting Unclean comments.

Transfer learning Binary Classification

The model was trained with 2 epochs, but it was computationally expensive.



Metric	Value
Threshold	0.86
Test ROC-AUC	0.973
Test accuracy	0.920
Test precision	0.554
Test recall	0.905
Test F1	0.687

Transfer learning Binary Classification

Binary Transfer Learning Test Confusion Matrix

	Predicted clean	Predicted unclean
Actual clean	53,196	4,539
Actual unclean	596	5,647

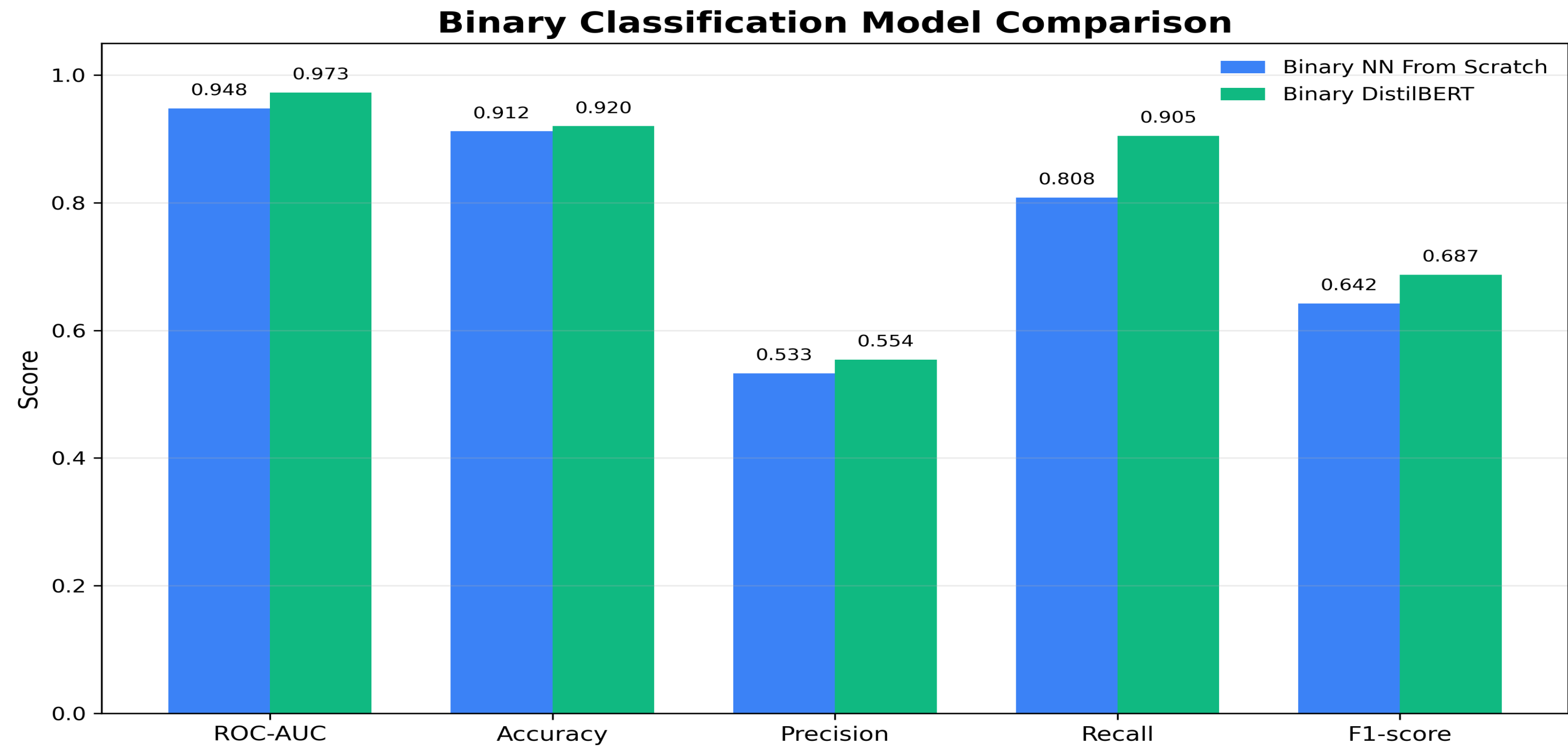
Green cells are correct decisions. Yellow/red cells are review errors.

Binary Transfer Learning Validation Confusion Matrix

	Predicted clean	Predicted unclean
Actual clean	28,306	364
Actual unclean	599	2,646

Green cells are correct decisions. Yellow/red cells are review errors.

Binary Classification



TOXIC COMMENT CLASSIFICATION PROJECT

Thank You

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